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Microsphere-based insulating materials for use in vacuum insulated structures

Abstract

A low-density insulating material for use in a vacuum insulated structure for an appliance includes a plurality of microspheres that includes a plurality of leached microspheres. Each leached microsphere has an outer wall and an interior volume. The outer wall has a hole that extends through the outer wall and to the interior volume. A binder engages outer surfaces of the plurality of leached microspheres, wherein the binder cooperates with the plurality of leached microspheres to form at least one microsphere aggregate. The interior volume of each leached microsphere defines an insulating space that includes an insulating gas. The insulating space of each leached microsphere is at least partially defined by the binder.

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Background/Summary

FIELD OF THE DEVICE

(1) The device is in the field of insulating materials for use in insulating structures, and more specifically, a microsphere-based insulating system that can be disposed within an insulating structure for an appliance and other vacuum-insulated structures.

SUMMARY

(2) In at least one aspect, a low-density insulating material for use in a vacuum insulated structure for an appliance includes a plurality of microspheres having a plurality of leached microspheres. Each leached microsphere has an outer wall and an interior volume. The outer wall has at least one hole that extends through the outer wall and to the interior volume. A binder engages outer surfaces of the plurality of leached microspheres, wherein the binder cooperates with the plurality of leached microspheres to form at least one microsphere aggregate. The interior volume of each leached microsphere defines an insulating space that includes an insulating gas. The insulating space of each leached microsphere is at least partially defined by the binder.

(3) In at least another aspect, a refrigerating appliance includes an inner liner and an outer wrapper that define an insulating cavity therebetween. An at least partial vacuum is defined within the insulating cavity. A plurality of microsphere aggregates are disposed within the insulating cavity. Interstitial spaces are defined between the microsphere aggregates. Each microsphere aggregate includes at least one leached microsphere having a leached opening that extends through an outer wall and to an interior volume of each leached microsphere. A binder engages an outer surface of the outer wall, wherein the binder is disposed within a portion of the leached opening and at least partially defines the interior volume. The interior volume of each leached microsphere defines an insulating space that is at least partially defined by the binder. The binder and the outer wall define a substantially air-tight seal around the insulating space.

(4) In at least another aspect, a method for forming an insulating microsphere for use in an appliance cabinet includes heating glass particles to a predetermined temperature to define molten glass particles, wherein the molten glass particles form a microsphere having an outer wall and an interior volume that defines a hollow microsphere. The molten glass particles are cooled to an intermediate temperature, to define solid microspheres, wherein the intermediate temperature is less than approximately 400 degrees Celsius. The solid microspheres are coated with an opacifier, wherein the heating, cooling and coating steps occur within a single assembly.

(5) In at least another aspect, a method for forming low-density microsphere aggregates includes leaching a plurality of microspheres to define leached microspheres, wherein each leached microsphere includes an outer wall and an interior volume. Each leached microsphere includes at least one of a cavity and a hole within the outer wall of the leached microsphere. The method also includes evacuating air from within the interior volume of each leached microsphere having a hole to define an insulating space within each leached microsphere having a hole. The method also includes coating the leached microspheres with a binder, wherein the binder engages the leached microspheres and at least partially occupies at least a portion of the holes of the leached microspheres, and the binder at least partially defines the insulating space of the leached microsphere having a hole. The method also includes mixing the leached microspheres and the binder within a mixer. Mixing of the leached microspheres and the binder results in a plurality of microsphere aggregates, and at least a portion of the leached microspheres within the microsphere aggregates includes the insulating space.

(6) These and other features, advantages, and objects of the present device will be further

understood and appreciated by those skilled in the art upon studying the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In the drawings:
(2) FIG. **1** is a front perspective view of a refrigerating appliance that incorporates an aspect of the microsphere insulating material;
(3) FIG. **2** is a cross-sectional view of a wall for a cabinet of the appliance of FIG. **1**, taken along line II-II;
(4) FIG. **3** is a cross-sectional view of the appliance of FIG. **1** showing an aspect of the microsphere insulating material;
(5) FIG. **4** is an enlarged view of the microsphere insulating material of FIG. **3** taken at area IV and showing packed microspheres;
(6) FIG. **5** is an enlarged view of an aspect of the insulating material of FIG. **3** showing a combination of microspheres and additional particulate material disposed within an insulating cavity of the appliance;
(7) FIG. **6** is an enlarged view of an aspect of the insulating material of FIG. **3** and showing microspheres and insulating particulate material combined together within the insulating cavity;
(8) FIG. **7** is a schematic flow diagram illustrating an aspect of a method for forming an opacifier-coated microsphere;
(9) FIG. **8** is a schematic graph showing a process for forming the opacifier-coated microspheres during a cooling phase of the microsphere formation process;
(10) FIG. **9** is a schematic cross-sectional view of a microsphere aggregate that includes a plurality of microspheres held together by a binder;
(11) FIG. **10** is a schematic diagram illustrating a process for forming leached microspheres;
(12) FIG. **11** is a schematic graph showing the use of heat in forming leached microspheres that incorporates a cooling phase of the microsphere formation process;
(13) FIG. **12** is a schematic diagram illustrating an assembly used for forming microspheres that include an at least partial vacuum within an interior volume of the microspheres;
(14) FIG. **13** is a schematic diagram illustrating an assembly for forming microspheres that include an inert gas contained within the interior volume of the microsphere;
(15) FIG. **14** is a schematic diagram illustrating a method for forming the microsphere aggregates using leached microspheres;
(16) FIG. **15** is a schematic cross-sectional view of a leached microsphere that is coated with a binder and includes a controlled environment within the interior volume of the microsphere;
(17) FIG. **16** is a schematic flow diagram illustrating a method for forming opacifier coated insulating microspheres for use in an appliance cabinet;
(18) FIG. **17** is a linear flow diagram illustrating a method for forming a low density microsphere-based insulating material;
(19) FIG. **18** is a linear flow diagram illustrating a method for forming a low density microsphere-based insulating material; and
(20) FIG. **19** is a schematic diagram illustrating performance of a leaching process using a mesh container within an acid bath.
DETAILED DESCRIPTION OF EMBODIMENTS
(21) For purposes of description herein the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the device as oriented in FIG. **1**. However, it is to be understood that the device may assume various alternative orientations and

step sequences, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(22) With respect to FIGS. **1**, **2**, **9** and **15**, reference numeral **10** generally refers to a low-density insulating material that can be disposed within an insulating cavity **12** of an appliance **14**. The refrigerating appliance **14** includes a cabinet **16** that forms the structure for the appliance **14**. The cabinet **16** can include an outer wrapper **18** and an inner liner **20** that cooperate to form the insulating cavity **12** therebetween. The cabinet **16** of the appliance **14** can define a vacuum insulated structure **22** where air is expressed, expelled, or otherwise removed from the insulating cavity **12** of the cabinet **16** to form an at least partial vacuum **26** within the vacuum insulated structure **22**. In various aspects of the device, the vacuum insulated structures **22** can be pre-formed and then placed within the insulating cavity **12** to increase the thermal insulating properties of the appliance **14**. The appliance **14** can include various doors **28** and drawers **30** that can be used to provide access to various interior compartments **32** of the appliance **14**.

(23) As exemplified in FIGS. **1**, **2**, **9** and **15**, the refrigerating appliance **14** can include the inner liner **20** and outer wrapper **18** that define the insulating cavity **12** therebetween, and wherein an at least partial vacuum **26** is defined within the insulating cavity **12**. A plurality of microsphere aggregates **34** are disposed within the insulating cavity **12** and various interstitial spaces **36** are formed between the microsphere aggregates **34**. In various aspects of the device, a secondary insulating material **38**, in the form of an insulating powder, insulating microspheres **52**, silica-based insulating materials, combinations thereof and other similar insulating components can be disposed within these interstitial spaces **36** between the microsphere aggregates **34**.

(24) Referring again to FIGS. **1**, **2**, **9** and **15**, these microsphere aggregates **34** can include one or more leached microspheres **40** having a leached opening **42** that extends through an outer wall **44** of the leached microsphere **40** and into the interior volume **46** of the leached microsphere **40**. These leached microspheres **40** can be made to include at least one cavity **48** and through holes **50** that are formed through a process of leaching by heating glass microspheres **52** to a particular temperature and then placing the heated microspheres **54** (shown in FIG. **10**) into an acid bath **56**. In the acid bath **56**, various components of the microspheres **52** can be removed from the outer wall **44** of the microsphere **52** and leaving the various cavities **48** and through holes **50**. The process of leaching will be described more fully below.

(25) Referring again to FIGS. **1**, **2**, **9** and **15**, the microsphere aggregates **34** can also include a binder **70** that engages an outer surface **72** of the outer wall **44** of the plurality of leached microspheres **40**. The binder **70** cooperates with the plurality of leached microspheres **40** to form at least one microsphere aggregate **34**. The binder **70** can be disposed within a portion of the various leached openings **42** in the form of the cavities **48** and through holes **50** defined within the outer wall **44** of the various leached microspheres **40**. In this manner, the binder **70** that is disposed within these leached openings **42**, and particularly within the through holes **50**, can at least partially define the interior volume **46** of the various leached microspheres **40**. In various aspects of the device, the interior volume **46** of each leached microsphere **40** of the microsphere aggregates **34** defines an insulating space **74**. This insulating space **74** can be at least partially defined by the binder **70** that occupies the various leached openings **42** of the leached microspheres **40** of the microsphere aggregates **34**. With the binder **70** surrounding the outer wall **44** and the various leached openings **42** of the leached microspheres **40**, the binder **70** and the outer wall **44** cooperate to define a substantially air-tight seal around the insulating space **74** of the leached microsphere **40**. This insulating space **74** can be defined by an at least partial vacuum **26** where air **24** is evacuated from the interior space **262** of the microsphere **52** and a low pressure area is defined within this insulating space **74**. Various insulating gases **76** can also be disposed within the insulating space

74. These insulating gases **76** can be maintained at atmosphere or could be maintained at a lower pressure. Accordingly, a partial vacuum **26**, in combination with the insulating gas **76**, can define the insulating space **74** of the various leached microspheres **40** within the microsphere aggregates **34**.

(26) Referring again to FIG. **2**, within the cabinet **16** for the appliance **14**, the insulating cavity **12** defined between the inner liner **20** and outer wrapper **18** can hold the microsphere aggregates **34** as part of the low-density insulating material **10**. Along with the microsphere aggregates **34**, the insulating cavity **12** of the cabinet **16** can include an insulating gas **76** that is disposed within the insulating cavity **12** to substantially occupy the interstitial spaces **36** and/or various secondary interstitial spaces **78**. These secondary interstitial spaces **78** are defined between the plurality of microsphere aggregates **34** and also between various particles of insulating powder and other components of a secondary insulating material **38** that are disposed within the insulating cavity **12** having the microsphere aggregates **34**. By using the microsphere aggregates **34**, the secondary insulating material **38** and the insulating gas **76**, the entire insulating cavity **12**, or substantially all of the insulating cavity **12**, can be filled with various insulating components that improve the thermal insulating properties of the cabinet **16** and other panel components for the appliance **14**.

(27) According to various aspects of the device, as exemplified in FIG. **2**, the microsphere aggregates **34** can substantially fill the insulating cavity **12** defined within the cabinet **16** for the appliance **14**. The microsphere aggregates **34** provide an interior structure **90** within the insulating cavity **12** that supports the positioning of the inner liner **20** and the outer wrapper **18** with respect to one another. In various aspects of the appliance **14**, air is drawn from the insulating cavity **12** to define an at least partial vacuum **26** within the insulating cavity **12**. During this process of expressing, expelling, or otherwise removing air **24** from the insulating cavity **12**, the resulting partial vacuum **26** within the insulating cavity **12** results in inward compressive forces **92** being exerted against an exterior surface **94** of the cabinet **16** along the outer wrapper **18** and the inner liner **20**. The plurality of microsphere aggregates **34** disposed within the insulating cavity **12** engage the inner surface **96** of the cabinet **16** that defines the insulating cavity **12**. This positioning of the plurality of microsphere aggregates **34** serves to define the interior structure **90** that substantially opposes the inward compressive force **92** exerted upon the exterior surface **94** of the cabinet **16** for the appliance **14**. Through the use of the interior structure **90** made up of the microsphere aggregates **34**, when the vacuum is defined within the insulating cavity **12**, the microsphere aggregates **34** serve to resist inward deflection of the inner liner **20** and outer wrapper **18** that may otherwise result from the inward compressive force **92**. Accordingly, various deformation and other types of aesthetic demarcation within the exterior surface **94** of the inner liner **20** and the outer wrapper **18** can be substantially prevented. This deflection caused by the inward compressive force **92** of the partial vacuum **26** can also lead to thinning of various portions of the cabinet **16** and diminished thermal performance within these areas. The use of the microsphere aggregates **34** serves to oppose this deformation such that the cabinet **16** defines a substantially consistent width **98** throughout the various walls of the structural cabinet **16** for the appliance **14**.

(28) To provide the interior structure **90** defined by the microsphere aggregates **34**, it is typical that each microsphere aggregate **34** of the plurality of microsphere aggregates **34** is in direct engagement with at least one adjacent microsphere aggregate **110** of the plurality of microsphere aggregates **34**. In this manner, the various microsphere aggregates **34** support one another within the insulating cavity **12** to prevent the deformation or aesthetic demarcation within the structural cabinet **16**. As discussed previously, the microsphere aggregates **34**, when disposed and packed within the insulating cavity **12**, define various interstitial spaces **36** therebetween. These interstitial spaces **36** can be occupied by secondary insulating materials **38** that define secondary interstitial spaces **78**. An insulating gas **76** can be used to substantially occupy the interstitial spaces **36** and the secondary interstitial spaces **78**. In various aspects of the device, the interstitial and secondary

interstitial spaces **36**, **78** can be maintained at an at least partial vacuum **26**.
(29) Referring again to FIGS. **2**, **9** and **15**, the various microsphere aggregates **34** can be surrounded by an opacifier **120** that substantially coats each microsphere aggregate **34**. According to various aspects of the device, the opacifier **120** can be disposed within the binder **70**. The opacifier **120** can also be delivered by the binder **70** during formation of the microsphere aggregates **34**.
(30) According to various aspects of the device, the opacifier **120** that is disposed within or around the microsphere aggregate **34** can be disposed proximate or around an outside surface **290** or at least a portion of the outside surface **290** of each microsphere aggregate **34**. In such an embodiment, it is typical that the microsphere aggregates **34** are formed and an opacifier **120** is placed around the microsphere aggregates **34**. As discussed previously, the binder **70** can contain or at least partially deliver portions of the opacifier **120** that make up the components of the microsphere aggregate **34**.
(31) According to various aspects of the device, the binder **70** used to form the microsphere aggregates **34** can be in the form of polyethylene glycol (PEG), resin, one or more natural waxes, one or more synthetic waxes, combinations thereof, and other similar materials that can be used to solidify and bind together various microspheres **52** to form the microsphere aggregates **34**.
(32) According to various aspects of the device, the opacifier **120** that is used to coat or at least partially coat the various microsphere aggregates **34** can be in the form of carbon black, silicon carbide, zinc oxide, titanium oxide, rice husk ash, other plant material having a high silica content, combinations thereof, and other similar materials having a high resistance to radiative thermal conductivity.
(33) Additionally, the various insulating gases **76** used within the formation of the microspheres **52** and in the formation of the structural cabinet **16** can include, but are not limited to, carbon dioxide, argon, xenon, krypton, neon, combinations thereof, and other similar thermally insulating gases **76**.
(34) The secondary insulating material **38** that can be disposed within the interstitial spaces **36** defined between the microsphere aggregates **34** can include, but is not limited to, various opacifiers **120**, silica-based materials, insulating powders, microspheres **52**, hollow microspheres, glass fiber, insulating gas **76**, combinations thereof, and other similar insulating materials.
(35) Referring now to FIGS. **9-11** and **15**, the leached microspheres **40** have various portions of the outer wall **44** of the microspheres **52** removed therefrom. The removal of this material is typically a result of the leaching process **130**. In the leaching process **130**, the microspheres **52** are heated to a phase separating temperature **132** within a range of from approximately 400 degrees Celsius to approximately 900 degrees Celsius. In this heating process **190**, which can last from approximately one half hour to approximately two hours, the microspheres **52** experience a phase separation where certain materials within the microspheres **52** migrate to an outer periphery **134** of the microsphere **52**. Typically, the leeched microspheres **40** and the partially leached microspheres **150** are derived from boron-based glass. The boron-based glass can include, but is not limited to, borosilicate glass, aluminosilicate glass, boroaluminosilicate glass and other similar glass-type materials. During the leaching process **130**, the boron **136** within the borosilicate glass migrates to the outer periphery **134** of the microsphere **52**. After the heating process **190** is complete, the phase separated microspheres **138** are placed within an acid bath **56** and moved within the acid bath **56**. Typically, hydrochloric acid is used as the acid, however, other acids can be used for interacting with the migrated boron **136**. Because the boron **136** has migrated to the outer periphery **134** of the microsphere **52**, the acid interacts with the boron **136** and leaches and/or etches the boron **136** away from the outer wall **44** of the microsphere **52**. The remaining material, typically a silica-based material, defines an outer wall **44** that includes the cavities **48** and through holes **50** where the boron **136** has been removed. The acid does not typically interact with the silica-based material.
(36) Additionally, in the heating process **190**, the temperature experienced by the heated and phase-separated microspheres **54**, **52** causes the boron **136** to migrate to the outer periphery **134** of the

microsphere **52**. The remaining components of the microsphere **52** remains substantially intact. As will be described more fully below, the leached microspheres **40** can be used to remove air **24** from, or to inject gas into, the insulating space **74** defined within the interior volume **46** of the leached microspheres **40**.

(37) By using leached microspheres **40** that have boron **136** removed, the microsphere aggregates **34**, having the leached microspheres **40**, are much lighter and less dense than aggregates that include intact microspheres **52**. Accordingly, the use of the microsphere aggregates **34** can be used to form a low-density insulating material **10**. Because the overall weight of each microsphere aggregate **34** is less, the overall weight of the low-density insulating material **10** is also less, and, in turn, the overall weight of the appliance **14** is also decreased through the use of the low-density insulating material **10**.

(38) Through the leaching process **130**, it should be understood that certain microspheres **52** may not be fully leached such that some amount of boron **136** may remain within various leached microspheres **40**. Additionally, some of the leached microspheres **40** may also be free of through holes **50**. These partially-leached microspheres **150** (shown in FIG. **14**) may contain cavities **48** (shown in FIG. **15**) within the outer wall **44** that do not extend entirely through the outer wall **44** of the microsphere **52**. In these partially-leached microspheres **150**, the interior volume **46** of the microsphere **52** will typically contain gas in the form of air **24**. In various aspects of the device, the interior volume **46** of the partially-leached microsphere **150** may also include an insulating gas **76** in the form of carbon dioxide or other byproduct generated during the formation of the microsphere **52**. This process will be described more fully below.

(39) In various aspects of the device, individual microspheres **52** can also be used within the insulating cavity **12** in the form of the secondary insulating material **38**. These individual microspheres **52** can also be manufactured within a single assembly so that various sequential steps can be performed in preparing a microsphere-based insulating material **180** for use in an insulating cavity **12** of an appliance **14**.

(40) As exemplified in FIGS. **3-8** and **16**, microspheres **52** can be formed and coated with an opacifier **120** within a single assembly. Through this assembly, the various microspheres **52** can be formed from glass particles **160** or glass frit and, during a cooling phase **162** of the glass microspheres **52**, can be coated with an opacifier **120**. As discussed above, the opacifier **120** used typically reduces the radiative component of thermal conductivity of the article that it surrounds, in this case, the microspheres **52**. As exemplified in FIG. **16**, a method **400** is disclosed for forming an insulated microsphere **52** in the form of an opacifier-coated microsphere **164** for use in an appliance cabinet **16**. This method **400** can include step **402** of heating glass particles **160** to a predetermined temperature **166** to define molten glass particles **168**. Typically, the glass particles **168** are pre-sized glass particles **168** that are designed to result in a microsphere **52** having a correspondence size. To form the glass microspheres **52**, the glass particles **160** are heated to the predetermined temperature **166** of from approximately 900 degrees Celsius to approximately 1100 degrees Celsius. This predetermined temperature **166** creates molten glass particles **168** that generate a gas within the interior of the molten glass particles **168**. As this gas is generated, the molten glass particle **168** takes the form of a generally spherical shape having an outer wall **44** with an interior volume **46** defined therein. This generally spherical shape having the interior volume **46** is generally described as a hollow microsphere **52**. The molten glass particles **168** are then cooled during a cooling phase **162** to an intermediate temperature **170** where the various components of the molten glass particles **168** can solidify into heated microspheres **54** (step **404**). The temperature that the components of the molten glass particles **168** solidify can be approximately 400° C. Once this intermediate temperature **170** is reached, the heated solid microspheres **172** can typically be coated with an opacifier **120** (step **406**). The components used for the opacifier **120** can be those discussed previously. The opacifier **120** can be adhered to the heated solid microspheres **172** through static adhesion. A binder or other adhering material can also be used to adhere the opacifier

120 to the heated solid microspheres **172**.

(41) Referring again to FIGS. **3-8** and **16**, the time period used to create the opacifier-coated microspheres **164** can be relatively short. Formation of the molten glass particles **168** may take only a fraction of a second. Because the molten glass particles **168** are so small, the cooling phase **162** of the molten glass particles **168** can be relatively quick and in the order of minutes. During this cooling phase **162**, the various heated solid microspheres **172** can be placed within a drum and agitated within the drum where an opacifier **120** is also disposed therein. While being mixed within the drum, and opacifier **120** and the heated solid microspheres **172** can combine to form the opacifier-coated microspheres **164**. The microspheres **52** can be cooled using air, a heat exchanger, fluids, or other similar cooling process and/or medium. These opacifier-coated microspheres **164** can be formed during the cooling phase **162** of the various microspheres **52**. Accordingly, the overall time to create the opacifier-coated microspheres **164** can be less than ten minutes. Once the opacifier-coated microspheres **164** are generated, these opacifier-coated microspheres **164** can be moved through an assembly line for packaging, delivery, or installation within an insulating structure. By combining the processes of forming the microspheres **52** and coating the microspheres **52** into a single assembly, various steps in the process of forming an opacifier-coated microsphere **164** can be minimized and additional transportation steps can be made substantially unnecessary. This can save great expense and limit the use of resources during formation of the opacifier-coated microspheres **164**.

(42) Referring now to FIGS. **3-6**, the opacifier-coated microspheres **164** can be used within a microsphere-based insulating material **180** in various applications. The opacifier-coated microspheres **164** can be packed within an insulating cavity **12** (FIG. **4**). The interstitial spaces **36** or secondary interstitial spaces **78** between the opacifier-coated microspheres **164** can be depressurized to form an at least partial vacuum **26**. These interstitial spaces **36** can also or alternatively be at least partially filled with an insulating gas **76**. The opacifier-coated microspheres **164** can also be loosely packed with a secondary insulating material **38** as exemplified in FIG. **5**. The combination of the opacifier-coated microspheres **164** and the secondary insulating material **38** can form a substantially homogenous microsphere-based insulating material **180** that can be filled within an insulating cavity **12**. The opacifier-coated microspheres **164** can also be tightly packed with the secondary insulating material **38** as exemplified in FIG. **6**. The use of the opacifier-coated microspheres **164** can also be incorporated within the secondary insulating material **38** that is used in conjunction with the microsphere aggregates **34** discussed above. This secondary insulating material **38** can include the opacifier-coated microspheres **164** or can be added with another secondary insulating material **38** to be combined within the interstitial spaces **34** defined between the microsphere aggregates **34**.

(43) Referring now to FIGS. **9-11** and **17**, a method **500** is disclosed for forming leached microspheres **40** for use in a low-density insulating material **10**. As discussed above, leached microspheres **40** can be used within the microsphere aggregates **34**. A benefit of using the leached microspheres **40** is in the existence of the through holes **50** that extend through the outer wall **44** of the leached microsphere **40** so that air **24** and/or insulating gas **76** can be moved into or out of the interior volume **46** of the leached microsphere **40**. According to the method **500**, glass particles **160** are heated to the predetermined temperature **166** to define molten glass particles **168** (step **502**). As discussed above, this predetermined temperature **166** can be within the range of approximately 900° C. to approximately 1100° C. Through this heating process **190**, the molten glass particles **168** each define an outer wall **44** and an interior volume **46** defined therein in the general shape of a hollow microsphere **52**.

(44) According to the method **500**, the molten glass particles **168** are cooled during a cooling phase **162** to an intermediate temperature **170** to define a heated solid microsphere **172** (step **504**). Typically, this intermediate temperature **170** can have an upper limit of approximately 400° C. At this temperature, the components of the heated solid microsphere **172** are typically substantially

solidified. These heated solid microspheres **172** can then be reheated from the intermediate temperature **170** to a phase separating temperature **132** to define phase separated microspheres **138** (step **506**).

(45) As discussed above, the leached microspheres **40** are phase separated such that the boron **136** and/or other components of the microspheres **52** are heated and caused to migrate or otherwise move to the outer periphery **134** of the microsphere. Typically, this reheating process **194** to the phase separate the microspheres **52** and can last from approximately 30 minutes to approximately two hours.

(46) According to various aspects of the device, the phase separating temperature **132** can be in the range of approximately 400 degrees Celsius to 900 degrees Celsius. This temperature causes the boron **136** included within the borosilicate glass to become molten while the other components of the microsphere **52** remain substantially solidified. In this manner, the boron **136** migrates to the surface of the microsphere **52**. This configuration allows for migration of the boron **136** to the outer periphery **134** of each microspheres **52** so that the acid used in the acid bath **56** can access the boron **136**, as will be described more fully below.

(47) According to method **500**, the phase separated microspheres **138** are then leached within an acid solution in an acid bath **56** (step **508**). The acid bath **56** leeches boron **136** from the phase separated microspheres **138** to define the leached microspheres **40**. As discussed above, the phase separated microspheres **138** are defined by boron **136** being moved or migrating to the outer periphery **134** of each microsphere **52**. In this configuration, the acid within the acid bath **56** can conveniently access and interact with the boron **136** of the phase separated microsphere **138**. The boron **136** can then be leached, etched, eroded, or otherwise removed from the outer wall **44** of the microsphere **52**. Once the leaching process **130** is complete, the leached microspheres **40** typically include various cavities **48** and through holes **50** that are defined within and/or extend through the outer wall **44** of the microsphere **52**.

(48) Referring now to FIGS. **10**, **15** and **19**, the leaching process **130** can be performed within the acid bath **56** by placing the phase separated microspheres **138** within a mesh basket **300**. This mesh basket **300** can be made of a fine mesh having aperture sizes that are smaller than a typical diameter of the phase separated microspheres **138**. According to various aspects of the device, an exemplary mesh size can be formed into a 74 micron mesh container. This mesh container can include a structural cap **302** that encloses a portion of the mesh container. The structural cap **302** also has sufficient weight to cause the mesh container and the phase separated microspheres **138** contained therein to be fully submerged within the acid bath **56**. Typically, the phase separated microspheres **138** are hollow and will be generally buoyant within the acid bath **56**. If the phase separated microspheres **138** are allowed to float within the acid bath **56**, at least a portion of the phase separated microspheres **138** will float above the acid solution **304** and may avoid direct contact with the acid solution **304** disposed within the acid bath **56**. By using the mesh container having the structural cap **302**, the phase separated microspheres **138** contained therein can be submerged within the acid bath **56** so that the acid solution **304** within the acid bath **56** can engage substantially all of the outer surface **72** of the phase separated microspheres **138**.

(49) Referring again to FIGS. **10**, **15** and **19**, included within the structural cap **302** for the mesh container is a connector **306**, such as a hook, that can be used to attach a linkage **308** for raising and lowering the mesh container within the acid bath **56**. Once the leaching process **130** is complete, the linkage **308** can be operated to lift the mesh container, via the connector **306** of the structural cap **302**, out of the acid bath **56**. The mesh container has a mesh size sufficient to allow the acid solution **304** within the acid bath **56** to pass through the mesh of the mesh container and fill the contained space **310** within the mesh container.

(50) Referring again to FIGS. **10**, **15** and **19**, the structural cap **302** can also include a receiver **312** that links with a vibrating mechanism. According to various aspects of the device, this receiver **312** can engage a vibrating rod **314** that is connected with the structural cap **302**. The vibrating rod **314**,

in various aspects of the device, can extend through the structural cap **302** and extend into the contained space **310** of the mesh container. During operation of the leaching process **130**, the vibrating rod **314** can be coupled with the structural cap **302**. Activation of the vibrating rod **314** causes a vibrating action, such as an ultrasonic vibration **316**, that passes through the vibrating rod **314** and into the mesh container. This ultrasonic vibration **316** may also causes a vibration within the phase separated microspheres **138**. Ultrasonic vibration **316** of the phase separated microspheres **138** causes the phase separated microspheres **138** to vibrate against one another and prevent clumping or adhesion of the various phase separated microspheres **138**. By preventing clumping or adhesion of the various phase separated microspheres **138**, the acid solution **304** within the acid bath **56** can more efficiently surround each of the phase separated microspheres **138**. This can result in a more complete and homogenous leaching of the boron **136** from the phase separated microspheres **138**. Additionally, the use of the mesh basket **300**, the structural cap **301** and the ultrasonic vibration in conjunction with the acid bath **56** can produce leached microspheres **40** that have a greater number of holes **50**. Using this device in the leaching process **130**, the holes **50** also are smaller in size. Accordingly, the larger number of holes **50** provides better extraction of air and better infiltration of insulating gas **76** into the insulating space **74** of the leached microsphere **40**. Also, the smaller sized holes **50** result in a better configuration of the remaining silica material of the leached microsphere **40**. This configuration of the silica material provides a better structure and integrity for resisting the inward compressive forces **92**.

(51) According to various aspects of the device, the structural cap **302** can include a single receiver **312** that receives the linkage **308** and the vibrating bar within a single attachment of the structural cap **302**. In such an embodiment, the linkage **308** can deliver the ultrasonic vibration **316** to the structural cap **302** to be disseminated throughout the structural cap **302** of the mesh container and into the various phase separated microspheres **138** contained therein. In using the vibrating rod **314** with the mesh container, the ultrasonic vibration **316** can be a continuous vibration that occurs throughout the leaching process **130**. It is also contemplated that the ultrasonic vibration **316** can be an intermittent vibration or can be a vibration having a modified frequency throughout the leaching process **130**.

(52) Through the use of the mesh container and the structural cap **302** having the connector **306** and the receiver **312**, the use of the ultrasonic vibration **316** can be implemented to generate leached microspheres **40** having substantially homogeneity regarding the number of leached openings **42** defined within each leached microsphere **40** as well as the size of each of the leached openings **42** within the leached microspheres **40**. The use of the ultrasonic vibration **316** also provides for a more efficient leaching process **130** such that the acid solution **304** within the acid bath **56** can better engage the outer surface **72** of the phase separated microspheres **138**. Because the acid solution **304** has better access to each phase separated microsphere **138**, the leaching process **130** can be conducted in a faster and more efficient manner. The use of the mesh container having the structural cap **302** also allows for a user of such a device to be adequately shielded and/or separated from contact with the acid bath **56**.

(53) According to various aspects of the device, the mesh container can be made of various materials that can include, but are not limited to, plastic, metal, ceramic, composite, combinations thereof, and other similar materials that are resistant to the acid solution **304** contained within the acid bath **56**. The material used for the mesh container is also able to be formed into a wire mesh having a mesh size that is smaller than the diameter of the phase separated microspheres **138** and also smaller than the leached microspheres **40** that are formed as a result of the leaching process **130**. Typically, the mesh of the mesh basket **300** is a mesh size large enough to allow the leached boron **136** to pass out of the mesh basket **300**. As discussed above, the structural cap **302** of the mesh container is made of a material having a weight sufficient to fully submerge the mesh basket **300** and each of the phase separated microspheres **138** contained within the mesh container.

(54) According to various aspects of the device, the structural cap **302** can be made of a lighter

material and the linkage **308** that attaches to the structural cap **302** can push the mesh container and structural cap **302** downward to be fully submerged within the acid bath **56**. Typically, the structural cap **302** will have a sufficient weight to cause the submersion of the mesh container and the phase separated microspheres **138** within the acid bath **156**.

(55) As discussed above, the use of the ultrasonic vibration **316** transmitted through the vibrating rod **314** and into the structural cap **302** allows the powder contents that make up the phase separated microspheres **138** to remain broken apart such that no formation of agglomerates or aggregates occurs. By preventing the formation of agglomerates and aggregates, the acid solution **304** is able to be intermingled within and around each of the phase separated microspheres **138** so that the outer surfaces of each of the phase separated microspheres **138** can be engaged by the acid solution **304** of the acid bath **56** to quickly and efficiently complete the leaching process **130**.

(56) As part of the leaching process **130**, the phase separated microspheres **138** are cooled during a secondary cooling phase **210** to a leaching temperature **212** before they are included within the acid bath **56**. This leaching temperature **212** is typically within a range of from approximately 80 degrees Celsius to 100 degrees Celsius. This temperature maximizes the leaching process **130** so that the leached microspheres **40** have a sufficient amount of boron **136** removed from the outer wall **44** of the microsphere **52** to form the through holes **50** that provide access to the interior volume **46** of each microsphere **52**. Additionally, boron **136** is generally a heavy material in relation to the other components of the hollow microspheres **52**. With the boron **136** removed, the microspheres **52** are diminished in weight so that the overall weight of the microspheres **52** and the ultimate insulating material is much lighter than insulating materials that include un-leached microspheres **52**.

(57) According to method **500**, the leached microspheres **40** are re-cooled to the leaching temperature **212** (step **510**). In various aspects of the device, during this secondary cooling phase **210**, the leached microspheres **40** can be coated with an opacifier **120**. The step of coating the leached microspheres **40** with an opacifier **120** may also be omitted where the leached microspheres **40** will be added to a mixer **250** and blended with a binder **70**. In such an embodiment, the step of coating with an opacifier **120** can occur after the microsphere aggregates **34** are formed. Through the method **500**, the leached microspheres **40** can be formed within a relatively short period of time without fully cooling the formed leached microspheres **40**. During the process of forming the leached microspheres **40**, the glass particles **160** are heated to form the molten glass particles **168** and are cooled. Before the heated solid microspheres **170** are fully cooled, they are reheated to the phase separating temperature **132**. The leaching process **130** can last from approximately one hour to approximately three hours, depending upon various factors that can include, but are not limited to, the size of the microspheres **52**, the boron **136** content of the glass particles **160**, the amount of glass particles **160** to be leached, and other similar factors. In various aspects of the device, the leached microspheres **40** can be packed with an opacifier **120** and/or a secondary insulating material **38** in the absence of a binder **70**. In such an aspect, the leached microspheres **40** can allow for better packing due to the presence of the leached openings **42**. Additionally, the opacifier **120** and the secondary insulating material **38** can be packed around the leached microspheres **40** and can also be packed within the leached openings **42**. This can result in a more dense insulating material that is efficiently packed and includes an improved resistance to radiative thermal conductivity.

(58) According to the various aspects of the method **500**, a coating step **512** can include coating the leached microspheres **40** and/or the microsphere aggregates **34** with an opacifier **120**. In such an embodiment, the various steps and processes of the method **500** can occur within a single assembly. A single assembly can be in the form of an assembly line or various sequential mechanisms that can be combined to perform sequential steps that form the microspheres **52**, cool the microspheres **52**, leech the microspheres **52**, and then potentially coat the microspheres **52**. This assembly can be used to form leached microspheres **40** that can be moved to a separate location for delivery,

packaging, or installation within an insulating structure.

(59) Referring now to FIGS. **12-15** and **18**, having described various aspects of the low-density microsphere aggregates **34** and their inclusion within various insulating cabinets **16**, a method **600** is disclosed for forming a low-density microsphere aggregate **34**. According to the method **600**, a plurality of microspheres **52** are leached to define leached microspheres **40** (step **602**). As discussed above, the various leached microspheres **40** include an outer wall **44** that defines an interior volume **46** therein. Each leached microsphere **40** typically includes leached openings **42** in the form of cavities **48** and/or through holes **50** within the outer wall **44** of the leached microsphere **40**. As discussed above, the formation of the through holes **50** within the leached microspheres **40** allows for air **24** and various insulating gases **76** to be delivered into or removed from the interior volume **46** of each leached microsphere **40**.

(60) According to the method **600**, step **604** includes evacuating air **24** from within the interior volume **46** of each leached microsphere **40** having a through hole **50** within the outer wall **44**. By removing air **24** from the interior volume **46**, an insulating space **74** is defined within each leached microsphere **40** that includes one or more through holes **50**. As part of this step **604**, the evacuation of air **24** can be contemporaneously coincided with adding an insulating gas **76** to the interior volume **46** of the leached microsphere **40**.

(61) According to the method **600**, the leached microspheres **40** are coated with a binder **70**. The binder **70** during this step **606** engages the leached microsphere **40** and at least partially occupies at least a portion of the through holes **50** of the leached microspheres **40**. In this manner, the binder **70** at least partially defines the insulating space **74** of the leached microsphere **40** having one or more through holes **50**. Stated another way, the binder **70** forms a plug **240** (shown in FIG. **15**) that at least partially occupies the through hole **50**, and prevents gas from entering into or escaping from the insulating space **74** of the leached microsphere **40**. To better coat the leached microspheres **40** with the binder **70**, the leached microspheres **40** and the binder **70** are mixed within a mixer **250** (step **608**). The binder **70** is added to the leached microspheres **40** contemporaneously with operation of the mixer **250**. Mixing of the leached microspheres **40** and the binder **70** results in a plurality of microsphere aggregates **34**. At least a portion of the leached microspheres **40** within the microsphere aggregates **34** includes the insulating space **74**.

(62) According to the various embodiments, the size of the microsphere aggregates **34** can be dictated by the amount of binder **70** added and the mixing operation **252** performed by the mixer **250**. The mixing operation **252** can be defined by mixing speed, duration of the mixing operation **252**, configuration of the blade used within the mixer **250**, sequences of faster and slower mixing periods, periods of rest, combinations thereof and other similar mixing operations **252**. Typically, the microsphere aggregates **34** will be formed to have a size of from approximately 4 millimeters to approximately 0.1 millimeters. The microsphere aggregates **34** can also include varying shapes such as coral structures, strands, granules, irregular shapes, and other similar shapes. It should be understood that other sizes of microsphere aggregates **34** are possible depending upon the amount of binder **70** added and the type of mixing operation **252** performed as the binder **70** is added.

(63) Referring again to FIGS. **12**, **14** and **15**, the insulating space **74** of the leached microsphere **40** can be formed to define an at least partial vacuum **26** by performing the mixing operation **252** and adding the binder **70** within a vacuum chamber **260** that defines an at least partial vacuum **26**. The microspheres **52** can be added to the mixer **250**, and the air **24** can be expressed from the interior space **262** of the vacuum chamber **260**. By removing air **24** from the vacuum chamber **260**, air **24** is also removed through the interior volume **46** of the leached microspheres **40** via the through holes **50** of the outer wall **44**. Various sensors **264** can be added to an outlet **266** of the vacuum chamber **260** to assess how much air **24** or other gas remains within the vacuum chamber **260**. Once a desired degree of vacuum or pressurization is achieved, the operation for adding the binder **70** and the contemporaneous mixing operation **252** can be performed. Once this mixing operation **252** is complete and the appropriate amount of binder **70** is added, the microsphere aggregates **34**

are substantially formed and the insulating space **74** within each of the leached microspheres **40** is also formed. Because the binder **70** forms the plug **240** that surrounds and occupies at least a portion of the through holes **50**, air **24** is substantially prevented from infiltrating into this insulating space **74**. The level of vacuum desired within the insulating space **74** can be within a range of from approximately less than one millibar to approximately fifty millibars. Additionally, this level of vacuum within the insulating space **74** of the leached microspheres **40** can be maintained for an extended period of time.

(64) Referring now to FIGS. **13-15**, the insulating space **74** can also be defined by an insulating gas **76** that is maintained within the interior volume **46** of this microsphere **52**. In such an embodiment, the leached microspheres **40** can be disposed within a conditioned chamber **280** where an insulating gas **76** is injected into the conditioned chamber **280** via an inlet **268** and also removed from the conditioned chamber **280** via an outlet **266**. By filling the conditioned chamber **280** with the insulating gas **76**, the various interior volumes **46** defined within the leached microspheres **40** can also be occupied by the insulating gas **76**. In various aspects of the device, the insulating gas **76** can be disposed within the conditioned chamber **280** and the insulating gas **76** can also define an at least partial vacuum **26** such that the gas that is within the insulating chamber is in the form of the insulating gas **76**. In such an embodiment, more of the insulating gas **76** is removed via the outlet **266** than is injected into the conditioned chamber **280** via the inlet **268**. Accordingly, the insulating gas **76** can be defined within each of the insulating spaces **74** of the leached microspheres **40** and kept at a pressure similar to that of the at least partial vacuum **26** discussed above (i.e., approximately less than one millibar to approximately fifty millibars). During the mixing operation **252**, the insulating gas **76** is maintained therein and the binder **70** is added to the microspheres **52** and is subsequently or contemporaneously mixed so that the binder **70** surrounds the leached microspheres **40** and forms the plug **240** that substantially occupies portions of the through holes **50** extending therethrough. In this manner, the binder **70** prevents air **24** from entering into the insulating space **74** and also prevents the insulating gas **76** from leaving the insulating space **74**.

(65) According to various aspects of the device, after the microsphere aggregates **34** are formed, an opacifier **120** can be added to an outside surface **290** of the microsphere aggregates **34**.

(66) According to various aspects of the device, the microsphere aggregates **34** can be incorporated within the various insulating structures. These insulating structures can include, but are not limited to, vacuum insulated panels, vacuum insulated cabinets, structural cabinets for appliances **14** and other similar insulated structures. The microsphere aggregates **34** can also be used within various appliances **14**. These appliances **14** can include, but are not limited to, refrigerators, freezers, coolers, laundry appliances, dishwashers, hot water heaters, kitchen-type utensils, and other similar appliances **14** and fixtures.

(67) In the various mixing operations **252** defined herein, the mixing operation **252** can be performed by an impeller for a mixer **250**, a rotating drum, various rollers, agitating or vibrating mechanisms, particulate matter suspended within air **24** or an insulating gas **76** and delivered through a material to be coated or mixed, and other similar mixing, delivering, or agitating processes.

(68) According to various aspects of the device, as exemplified in FIGS. **7** and **10**, the insulating gas **76** defined within the interior volume **46** of the microspheres **52** or the insulating space **74** of the leached microspheres **40** can be in the form of carbon dioxide. In such an embodiment, the carbon dioxide can be formed therein as a byproduct of the formation of the microspheres **52**. When the glass particles **160** are first heated to form the molten glass particles **168**, combustion over or within a flame of the glass particles **160** can lead to formation of a carbon dioxide byproduct **296**. This carbon dioxide byproduct **296** can be entrapped within the interior volume **46** of the microsphere **52**. The various heating agents and additives can be added to the environment within which the glass particles **160** are heated to form the molten glass particles **168**. These additives and other elements can be used to increase the amount of carbon dioxide byproduct **296**

generated as a result of the formation of the microspheres **52**. This carbon dioxide byproduct **296** can be entrapped within the interior volume **46** of the microsphere **52**. In such an embodiment, where carbon dioxide byproduct **296** is disposed within a substantial part of the interior volume **46** of the microsphere **52**, the leaching process **130** may not be necessary for disposing the insulating gas **76** within the interior volume **46** or the insulating space **74** of the microsphere **52**. Various mechanisms and sensory equipment can be used to determine the contents of an interior volume **46** of the various microspheres **52**.

(69) It will be understood by one having ordinary skill in the art that construction of the described device and other components is not limited to any specific material. Other exemplary embodiments of the device disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.

(70) For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical, chemically or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.

(71) It is also important to note that the construction and arrangement of the elements of the device as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width **98** of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

(72) It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present device. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

(73) It is also to be understood that variations and modifications can be made on the aforementioned structures and methods without departing from the concepts of the present device, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

(74) The above description is considered that of the illustrated embodiments only. Modifications of the device will occur to those skilled in the art and to those who make or use the device. Therefore, it is understood that the embodiments shown in the drawings and described above is merely for illustrative purposes and not intended to limit the scope of the device, which is defined by the following claims as interpreted according to the principles of patent law, including the Doctrine of Equivalents.

Claims

1. A refrigerating appliance comprising: an inner liner and an outer wrapper that define an insulating cavity therebetween, wherein an at least partial vacuum is defined within the insulating cavity; a plurality of microsphere aggregates disposed within the insulating cavity, wherein interstitial spaces are defined between the microsphere aggregates, wherein each microsphere aggregate includes a size of from approximately 0.1 millimeters to approximately 4 millimeters, each microsphere aggregate comprising: at least one leached microsphere having a leached opening that extends through an outer wall and to an interior volume of each leached microsphere of the at least one leached microsphere; a binder that engages an outer surface of the outer wall, wherein the binder is disposed within a portion of the leached opening and at least partially defines the interior volume; wherein the interior volume of each leached microsphere defines an insulating space that is at least partially defined by the binder; the binder and the outer wall define a substantially air-tight seal around the insulating space.
2. The refrigerating appliance of claim 1, further comprising: an insulating gas that is disposed within the insulating cavity and substantially occupies the interstitial spaces.
3. The refrigerating appliance of claim 1, wherein the at least partial vacuum defined within the insulating cavity exerts an inward compressive force against exterior surfaces of the outer wrapper and the inner liner, and wherein at least the plurality of microsphere aggregates engage inner surfaces of the outer wrapper and the inner liner, wherein the plurality of microsphere aggregates defines an interior structure that substantially opposes the inward compressive force.
4. The refrigerating appliance of claim 3, wherein the interior structure defined by the plurality of microsphere aggregates substantially resists deflection of the outer wrapper and the inner liner toward one another.
5. The refrigerating appliance of claim 3, wherein each microsphere aggregate of the plurality of microsphere aggregates is in direct engagement with at least one adjacent microsphere aggregate of the plurality of microsphere aggregates.
6. The refrigerating appliance of claim 1, further comprising: an opacifier that coats the outer surface of the plurality of microsphere aggregates.
7. The refrigerating appliance of claim 1, wherein an opacifier is incorporated within the binder of the plurality of microsphere aggregates.
8. The refrigerating appliance of claim 1, wherein at least a portion of the plurality of microsphere aggregates includes at least one partially-leached microsphere, wherein each at least one partially-leached microsphere includes at least one cavity that extends partially into the outer wall.
9. The refrigerating appliance of claim 1, further comprising: an insulating powder disposed within the insulating cavity, wherein the insulating powder is disposed within the interstitial spaces defined between the microsphere aggregates.
10. The refrigerating appliance of claim 9, wherein an insulating gas is disposed within the insulating cavity and substantially occupies secondary interstitial spaces defined between the insulating powder and the plurality of microsphere aggregates.
11. The refrigerating appliance of claim 1, further comprising: an insulating gas that is disposed within the insulating space of at least a portion of the at least one leached microsphere.
12. The refrigerating appliance of claim 11, wherein the insulating gas includes at least one of carbon dioxide, argon, xenon, krypton, and neon.
13. The refrigerating appliance of claim 1, wherein the insulating space includes an at least partial vacuum.
14. The refrigerating appliance of claim 1, wherein the at least one leached microsphere is derived from a boron-based glass.
15. The refrigerating appliance of claim 14, wherein the boron-based glass includes at least one of

borosilicate glass, aluminosilicate glass, and boroaluminosilicate glass.
16. The refrigerating appliance of claim 1, wherein the binder includes at least one of polyethylene glycol, resin, natural wax, and synthetic wax.
17. The refrigerating appliance of claim 6, wherein the opacifier includes at least one of carbon black, silicon carbide, zinc oxide, rice husk ash, and titanium oxide.
18. The refrigerating appliance of claim 1, further comprising: a secondary insulating material disposed within the insulating cavity, wherein the secondary insulating material is disposed within the interstitial spaces defined between the plurality of microsphere aggregates.
19. The refrigerating appliance of claim 18, wherein the secondary insulating material includes at least one of an opacifier, a silica-based material, an insulating powder, microspheres, hollow microspheres, glass fiber, and an insulating gas.
20. The refrigerating appliance of claim 1, wherein the insulating spaces of the plurality of microsphere aggregates are maintained at a pressure similar to a pressure of the at least partial vacuum of the insulating cavity.